TOUR DE MATHS TRAINING for 1 June 2017

1.

m and *n* are integers with $m > n$. The number of integers between (but not including *m* and *n*) is

(a) $m - n$ (b) $m - n - 1$ (c) $m - n + 1$ (d) $m + n - 1$

2.

Successive discounts of 12% and 25% is the same as a single discount of

(a) 34% (b) 35% (c) 36% (d) 37%

3.

A and B run a race of $\frac{4}{5}$ km on a circular track having circumference $\frac{1}{4}$ km. They start together but run in opposite directions. If A wins by 40 metres, how far was B from the starting post when A passed it the first time?

(a) 12,5 metres (b) 15 metres (c) 17,5 metres (d) 20 metres

4.

The highest vehicle registration number (VRN) in Durban is just under ND 840000. The *sum total* of a VRN is the sum of the digits in the number, Thus the sum total of ND 678645 is

$6 + 7 + 8 + 6 + 4 + 5 = 36$.

How many VRNs in Durban have sum total greater that 49?

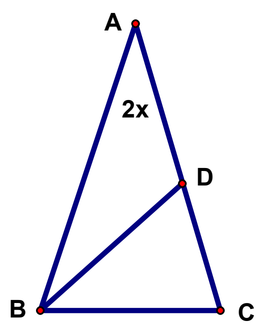
(a) 13 (b) 24 (c) 26 (d) 28

5.

In Δ ABC , AB = AC, BD = BC, and $\angle$ A = 2x.

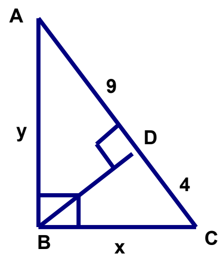
Then $\angle ABD$ is equal to

(a) $2x$ (b) $90^\circ - 3x$ (c) $90^\circ - 2x$ (d) $90^\circ - x$



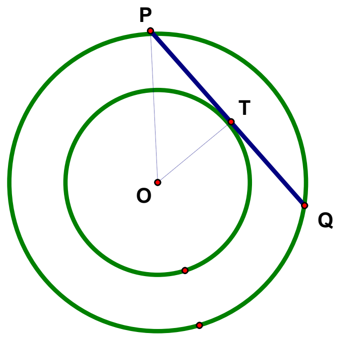
6.

$\angle ABC = \angle ADB = 90^\circ$ .What is the length of BD?



7.

The figure shows two concentric circles. If the area of the ring between the two circles is 144π square centimetres, what is the length of PQ, in centimetres?



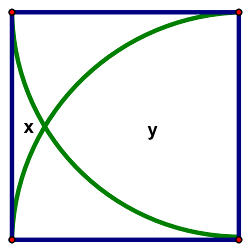
8.

When 333 333 333 333 333 333^2 + 222 222 222 222 222 222 is written out in full what is the sum of its digits?

9.

Four teams A, B, C and D each play a single game against each of the other teams. There are no draws and teams A, B C have the same number of wins. If team A beats team D, how many wins does team D have?

10.



The diagram shows a square and two quarter circles. Each quarter circle is centred on the vertex of the square. If a side of the square has length 2, and *x* and *y* are the areas of the indicated regions, what is *y* – *x* equal to?